04-27 第一篇

The Mystery of Black Holes: A Window into the Universe

Black holes are among the most puzzling objects in space. They form when a very large star runs out of fuel and collapses（坍塌） under its own gravity. [1] Nothing—not even light—can escape once it passes a point called the "event horizon（视界）." Scientists could not see black holes directly for a long time because they do not give off light. Instead, they study black holes by observing how they affect nearby stars and gas. In 2019, an international team released the first image of a black hole, which showed a bright ring around a dark center.

What happens inside a black hole? The truth is that nobody knows for sure. [2] The center of a black hole is called a "singularity（奇点）," where matter is squeezed into a point with almost no volume. Our current physics cannot fully explain what happens there. Some scientists think that black holes might connect to other parts of space-time through wormholes（虫洞）, though this remains only a theory. For now, black holes remain one of nature's deepest secrets.

Black holes are not just strange—they play an important role in the universe. [3] Supermassive black holes sit at the center of most galaxies, including our own Milky Way（银河系）. They help shape how galaxies form and grow over billions of years. When matter falls into a black hole, it heats up and sends out powerful radiation（辐射） before disappearing. By studying these processes, scientists learn more about how the universe works.

How do scientists find black holes if they cannot see them? [4] They watch for signs such as stars moving in unusual ways or X-rays coming from empty-looking regions of space. In 2015, scientists detected gravitational waves（引力波）—ripples in space-time—when two black holes merged（合并）. This opened a whole new way to study black holes and confirmed that Einstein's predictions from a century ago were correct.

[5] From invisible objects to the first photo, our understanding of black holes has grown rapidly. Yet there is still much to discover. As technology improves and new telescopes are built, we may one day unlock even more secrets of these fascinating（迷人的） cosmic（宇宙的） giants.

选项

A. In other words, black holes are central to the story of how galaxies—and perhaps life—came to exist. B. That is why scientists use indirect methods to detect and study them. C. In short, black holes have gone from pure mystery to something we can actually observe and measure. D. The gravity of a black hole is so strong that nothing can escape

from it. E. At this point, the laws of physics as we know them break down. F. Black holes are also thought to be key to understanding the birth of the universe.

答案键（Answer Key）

[1] D [2] E [3] A [4] B [5] C

深度解析

- **[1] D**：
  - **线索**：引出黑洞的基本特性——引力极强，任何事物都无法逃脱。空前说明"Black holes form when a very large star...collapses under its own gravity"，空后详述"Nothing—not even light—can escape once it passes...event horizon...Scientists could not see black holes directly...do not give off light"。D选项"The gravity of a black hole is so strong that nothing can escape from it"（黑洞的引力如此之强，以至于任何东西都无法逃脱）中"gravity...so strong...nothing can escape"与空后"Nothing—not even light—can escape"形成直接呼应，逻辑紧密，作为本段核心观点的概括句衔接顺畅。
- **[2] E**：
  - **线索**：黑洞内部与奇点的不可知性。空前提问"What happens inside a black hole? The truth is that nobody knows for sure"，空后说明"The center...called a 'singularity'...matter is squeezed into a point...Our current physics cannot fully explain what happens there...wormholes...remains only a theory"。E选项"At this point, the laws of physics as we know them break down"（在此处，我们所知的物理定律失效了）中"At this point"指代奇点，"laws of physics...break down"与空后"our current physics cannot fully explain"完美对应，衔接自然。
- **[3] A**：
  - **线索**：黑洞在宇宙中的重要作用。空前总述"Black holes are not just strange—they play an important role in the universe"，空后列举"Supermassive black holes sit at the center of most galaxies...help shape how galaxies form and grow...By studying these processes, scientists learn more about how the universe works"。A选项"In other words, black holes are central to the story of how galaxies—and perhaps life—came to exist"（换言之，黑洞在星系——或许还有生命——如何诞生的故事中扮演核心角色）中"In other words"对上文进行概括，"central to...how galaxies...came to exist"与空后具体描述形成呼应，衔接精准。
- **[4] B**：

- **线索**：科学家如何间接探测黑洞。空前提问"How do scientists find black holes if they cannot see them?"，空后详述"They watch for signs such as stars moving in unusual ways or X-rays...detected gravitational waves...This opened a whole new way to study black holes"。B选项"That is why scientists use indirect methods to detect and study them"（因此科学家使用间接方法来探测和研究它们）中"That is why"承接空前的"cannot see them"与"find"之间的矛盾，"indirect methods"与空后"watch for signs...gravitational waves"等间接手段完美对应，衔接顺畅。

- **[5] C**：
  - **线索**：全文收束与认知进步总结。空[5]位于结尾段段首，空后列举"From invisible objects to the first photo...understanding has grown rapidly...still much to discover...unlock even more secrets"。C选项"In short, black holes have gone from pure mystery to something we can actually observe and measure"（简言之，黑洞已从纯粹的谜团变成我们能够实际观察和测量的对象）用"In short"总结全文，"from pure mystery to...observe and measure"与空后"From invisible objects to the first photo...grown rapidly"形成升华，收束得当。

- **干扰项分析**：
  - **F. Black holes are also thought to be key to understanding the birth of the universe.**：强调黑洞对理解宇宙诞生的意义，易被误填[3]或[5]。但F聚焦"birth of the universe"，而空[3]前后文侧重"galaxies form and grow...how the universe works"，主题更宽泛，A的"how galaxies—and perhaps life—came to exist"更贴合；空[5]需总结人类认知进步（从不可见到可观测），F未涉及这一转变。F属"主题相关但范围与功能错位"型干扰。